

Rogers-Ramanujan Derivation — cycles 1-3

2026-05-15

Contents

Rogers-Ramanujan Derivation — cycles 1-3	1
Introduction	1
Definitions and Notation	1
Cycle 1: Formal Foundation and Finite Experiments	2
Cycle 2: Q-Difference Ladder and Finite Gap Recurrence	2
Cycle 3: Euler/Jacobi Product Transform	3
Current Proof Status	5
Open Questions	5
References	5
Appendix: Implementation Details	5

Rogers-Ramanujan Derivation — cycles 1-3

Introduction

Cycles 1-3 established a validated foundation for a first-principles proof of the two Rogers-Ramanujan identities, but did not complete the proof. The completed work produced formal-power-series definitions, reproducible finite experiments, proved series-side q-difference structure, proved finite gap recurrences, and proved the product-side Euler/Jacobi transformation infrastructure.

The current proof bottleneck is precise: the run has not yet proved that the Euler-multiplied Rogers-Ramanujan series collapse to the derived bilateral Jacobi sums. The auditor validated this as partial M2 progress and kept milestone M2 **in-progress** (auditor session 532d9472-a3f9-4ebd-ab75-90e005dc2cdf).

Definitions and Notation

The work is carried out in the formal power series ring $\mathbb{Z}[[q]]$. Equality means coefficientwise equality. The finite q-Pochhammer symbol is

$$(q; q)_n = \prod_{k=1}^n (1 - q^k), \quad (q; q)_0 = 1.$$

Cycle 1 introduced coefficient extraction $[q^k]A(q)$, congruence modulo q^{K+1} , and coefficientwise stabilization: a sequence $A_N(q)$ has a formal limit if each fixed coefficient eventually becomes constant. These definitions support all later finite-to-infinite passages (docs/lemmas/lemma_catalogue.md, worker session 8f4715a2-55b0-41e8-97b8-7a8ffd2c2709).

The two target series were also given a partition interpretation. The summand $q^{n^2}/(q; q)_n$ generates partitions into exactly n positive parts with adjacent gaps at least two, by subtracting the staircase $(2n - 1, 2n - 3, \dots, 1)$. The summand $q^{n^2+n}/(q; q)_n$ generates the analogous partitions with smallest part at least two, by subtracting $(2n, 2n - 2, \dots, 2)$.

Cycle 1: Formal Foundation and Finite Experiments

Cycle 1 built milestones M0 and M1. The researcher asked for a formal setup, finite coefficient harness, negative controls, and a validation document (84bce78a-c820-4739-950c-3445d9d2c1f0). The worker produced:

- docs/lemmas/lemma_catalogue.md
- scripts/rr/finite_rr_experiments.wls
- scripts/rr/plot_rr_difference_heatmap.py
- docs/validation.md
- coefficient, difference, recurrence, and figure outputs in data/finite_experiments/

The main finite experiment checked the two target series/products through degree 40. Both true pairs had no nonzero coefficient difference through that range. Negative controls failed early: residue classes $\{1, 3\}$ and $\{2, 4\}$ failed at degree 3, and the shifted exponent $n^2 + 2n$ failed at degree 1.

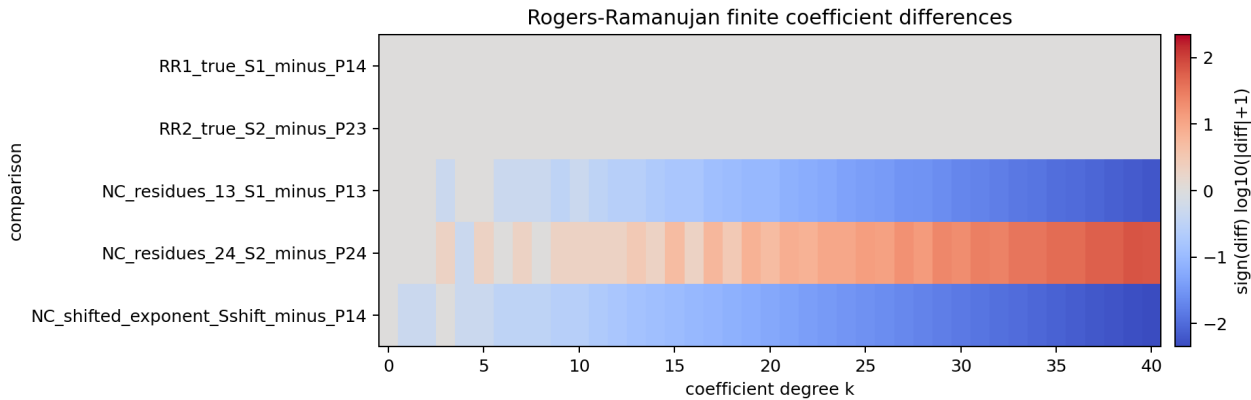


Figure 1: Heatmap of coefficient differences for the true residue products and negative controls. The true comparisons remain zero through the checked degree, while negative controls fail early.

The auditor validated M0 and M1 after inspecting the artifacts and rerunning a smaller Wolfram check. A ledger-format defect was repaired, but no mathematical defect was found (1759cc99-303e-4617-8ce0-56165df7f4f7).

Cycle 2: Q-Difference Ladder and Finite Gap Recurrence

Cycle 2 moved to M2: finding a proof-bearing transfer mechanism (bb2de806-9212-4328-8d2f-33a4f8f88a3c). The worker proved the auxiliary q-difference identity for

$$F(z, q) = \sum_{n \geq 0} \frac{z^n q^{n^2}}{(q; q)_n}.$$

The proved identity is

$$F(z, q) - F(zq, q) = zqF(zq^2, q).$$

Specializing $z = q^r$ gives the infinite ladder

$$A_r = A_{r+1} + q^{r+1}A_{r+2}, \quad A_r = F(q^r, q).$$

Here A_0 and A_1 are the two Rogers-Ramanujan series sides. The worker also proved tail-normalized uniqueness: there is at most one ladder solution satisfying $A_r \rightarrow 1$ coefficientwise as $r \rightarrow \infty$ (docs/proof/q_difference_mechanism.md, worker session 3503791b-e87a-459d-995e-a915715e61e2).

The same cycle proved a finite largest-part recurrence. If $G_N^{(a)}$ generates gap-two partitions with all parts at least a and largest part at most N , then

$$G_N^{(a)} = G_{N-1}^{(a)} + q^N G_{N-2}^{(a)}.$$

This comes from splitting partitions by whether the largest allowed part N appears (`docs/proof/finite_gap_recurrence.md`).

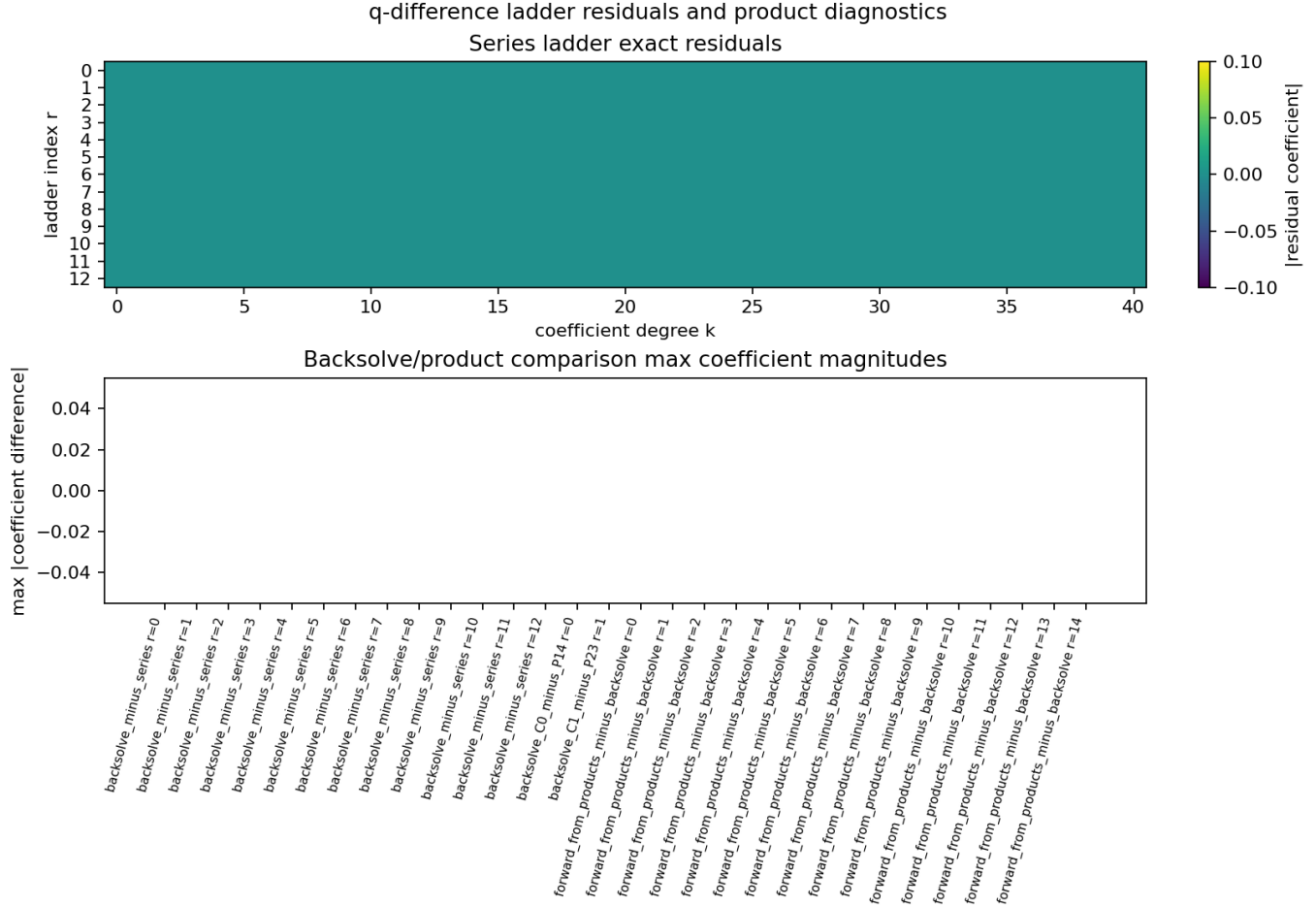


Figure 2: Residual plot for the q -difference ladder and backsolvable product comparisons. The series ladder residuals are zero through the checked range.

The auditor validated the cycle as partial M2 progress. The series-side ladder and uniqueness were accepted, but product-side closure was not proved. M2 remained `in-progress` (036adf0a-962e-4299-8aae-8ac788349d06).

Cycle 3: Euler/Jacobi Product Transform

Cycle 3 varied the M2 strategy toward an Euler-product and finite q -binomial route (221798d5-223f-4de3-8ca8-929e697ca61e). The worker proved that multiplying by

$$E(q) = (q; q)_\infty$$

reduces the target product identities to complementary modulo-5 products:

$$E(q)S_1 = (q^2; q^5)_\infty (q^3; q^5)_\infty (q^5; q^5)_\infty,$$

$$E(q)S_2 = (q; q^5)_\infty (q^4; q^5)_\infty (q^5; q^5)_\infty.$$

The worker then derived, rather than cited, a finite q-binomial theorem and a finite Jacobi-type identity. Passing coefficientwise to the limit gave

$$(z; Q)_\infty (Q/z; Q)_\infty (Q; Q)_\infty = \sum_{j \in \mathbb{Z}} (-1)^j z^j Q^{j(j-1)/2}.$$

With $(Q, z) = (q^5, q^2)$, this yields exponents $j(5j - 1)/2$. With $(Q, z) = (q^5, q)$, it yields $j(5j - 3)/2$. This proves the product/Jacobi half of the route (`docs/proof/product_transform_route.md`, worker session `f5efac66-0414-4e6e-a0c6-cd3f74a442df`).

The transformed finite series were

$$H_{\alpha, N} = \sum_{n=0}^N q^{n^2 + \alpha n} \frac{(q; q)_N}{(q; q)_n}.$$

They satisfy the proved recurrence

$$H_{\alpha, N} = (1 - q^N) H_{\alpha, N-1} + q^{N^2 + \alpha N}.$$

Experiments showed that the natural bilateral comparisons stabilize through degree N , but simple Schur-style Gaussian-window finite identities fail at degree 1. Therefore the transformed-series bilateral collapse remained conjectural as lemma L16.

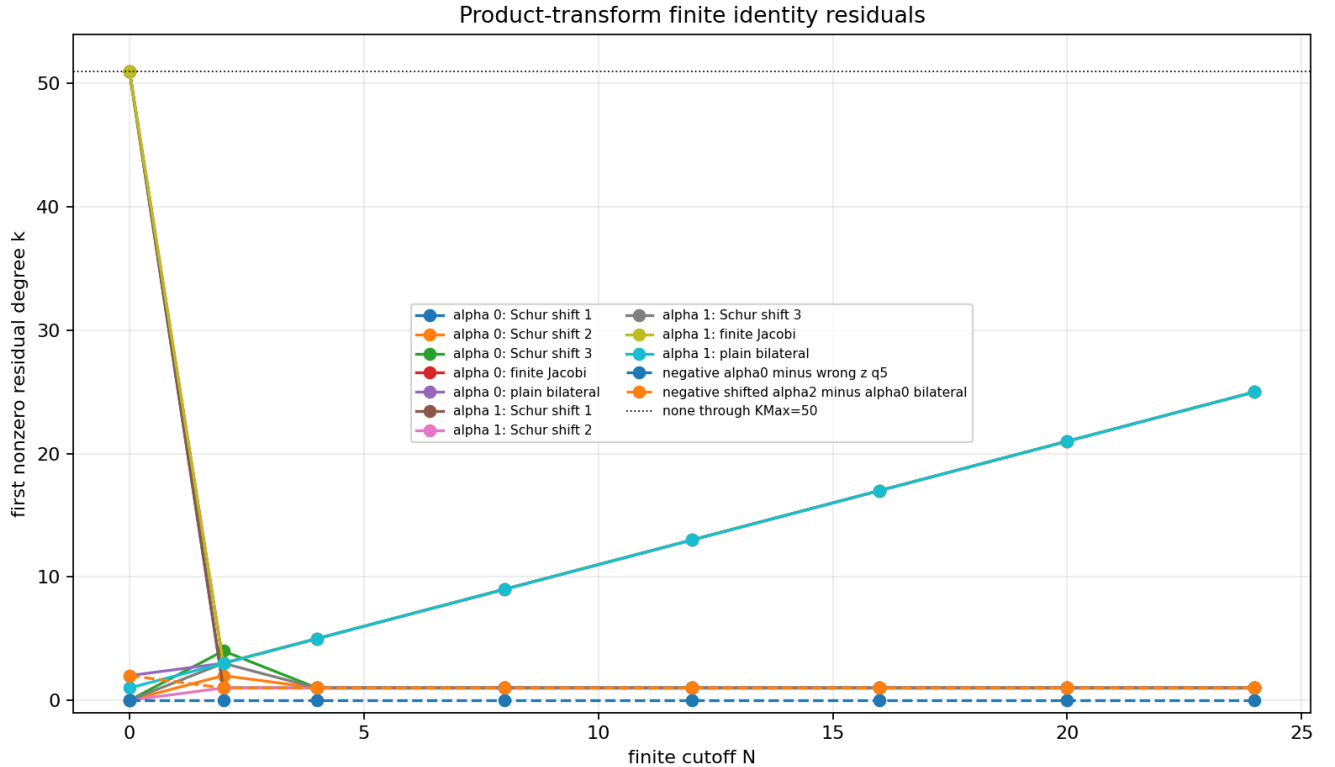


Figure 3: First-failure degree for transformed-series candidates and negative controls. Natural comparisons stabilize through degree N , while tested exact Gaussian-window formulas fail early.

The auditor validated the cycle as partial success. The plot script was fixed to support `--out`, ledger schema was repaired, and fresh Wolfram checks reproduced the stated residuals. The decision was **VALIDATED**, with M2 still open (`532d9472-a3f9-4ebd-ab75-90e005dc2cdf`).

Current Proof Status

Proved foundations:

- Formal coefficientwise equality, truncation transfer, and stabilization in $\mathbb{Z}[[q]]$.
- Staircase interpretations of both Rogers-Ramanujan series.
- Residue-product partition interpretation.
- Series-side q-difference ladder and tail-normalized uniqueness.
- Finite largest-part gap recurrence.
- Euler complement product reduction.
- Finite q-binomial theorem and derived Jacobi-type product identity.

Experimentally supported but not proved:

- The full Rogers-Ramanujan identities through finite coefficient checks.
- Product-side agreement with the q-difference backsolve.
- The transformed-series bilateral collapse $E(q)S_\alpha$ to the Jacobi bilateral sums.

Rejected or not sufficient:

- Naive bounded finite residue products as exact finite identities.
- Simple Schur-style Gaussian-window identities for $H_{\alpha,N}$.

Open Questions

The next cycle should target the missing transformed-series collapse:

$$E(q) \sum_{n \geq 0} \frac{q^{n^2 + \alpha n}}{(q; q)_n} = \sum_{j \in \mathbb{Z}} (-1)^j q^{j(5j - (2\alpha + 1))/2}, \quad \alpha = 0, 1.$$

The audit guidance recommends either a sign-reversing involution for coefficients of $(q; q)_N \sum_n q^{n^2 + \alpha n} / (q; q)_n$, or a richer finite state recurrence, likely involving modulo 5 or multiple coupled states.

References

No REFERENCES.md file exists in the workspace, and the cycle records state that no web search or external proof lookup was used. No external references are cited in this report.

Appendix: Implementation Details

Session references:

cycle	researcher	worker	auditor
1	84bce78a-c820-4739-950c-3945719a215b	3945719a-215b-b0-41e8-97b8-7b8f9cd2927309	7b8f9cd2-9273-09e-4617-8ce0-56165df7f
2	bb2de806-9212-4328-8d2f-33a4378f8e33	33a4378f-8e33-7a-459d-995e-a9167d15c619c2	a9167d15-c619-c2e-4299-8aae-8ac788349
3	221798d5-223f-4de3-8ca8-925e62766c14	925e6276-6c14-14-4e6e-a0c6-c532d94742a0ff	c532d947-42a0-ff9-4ebd-ab75-90e005dc2

Code organization:

path	files	role
scripts/rr/	7	Wolfram experiment harnesses and Python plotting scripts
docs/lemmas/	1	Lemma catalogue with proof statuses
docs/proof/	3	q-difference, finite-gap, and product-transform proof notes
data/finite_experiments/	21	CSV outputs and three figures

path	files	role
docs/validation.md	1	Separation of experiments, proof-supporting checks, and proved lemmas

Validation runs recorded across cycles include:

- `wolfram-batch -script scripts/rr/finite_rr_experiments.wls`
- `wolfram-batch -script scripts/rr/q_difference_probe.wls`
- `wolfram-batch -script scripts/rr/finite_gap_probe.wls`
- `wolfram-batch -script scripts/rr/product_transform_probe.wls`
- `python3 -m long_exposure.tools.promise_check <run-workspace>`
- `python3 -m long_exposure.tools.org_check <run-workspace>`

Current warnings are process/organization warnings only: M3 and M4 have no ledger events yet, `data/sessions.db*` are orphan artifacts, and `rogers_ramanujan_run_config.yaml` remains a root-file organization warning.

A workspace `MANIFEST.md` snapshot was updated for this report with script counts, line counts, and cross-references.